

TechCrush Capstone Project

PROJECT 2

Building a Secure Small Office Network

Network diagram, security controls, Packet Tracer simulation, threat response and reflection



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Group No. 6

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Outline

- Aim and Objective
- Concept of Project
- Network Design
- Devices and IP Address Plan
- Security Controls Implemented
- Methodology and Tools Used
- Packet Tracer Testing Evidence
- Simulated Threat and Response
- Results, Reflection and Conclusion
- References



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Aim and Objective

Aim

- Design and secure a basic small office network.
- Show how simple controls prevent guest users from reaching internal systems.
- Build confidence in explaining how networks are attacked and defended.

Objectives

- Draw a secure layout with router, firewall, switch, office devices, server and internet connection.
- Separate office and guest users into different network zones.
- Apply firewall and access-control rules to block guest access to the file server.
- Test and explain allowed office traffic and blocked guest traffic.



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Concept of Project

- The network is divided into a trusted Office Network and a lower-trust Guest Network.
- The File Server stays inside the office subnet as a protected internal resource.
- Guests can use internet access through the guest Wi-Fi but cannot reach office devices.
- A firewall traffic-control point enforces the separation between both zones.
- Packet Tracer tests confirm the expected allowed and blocked traffic paths.

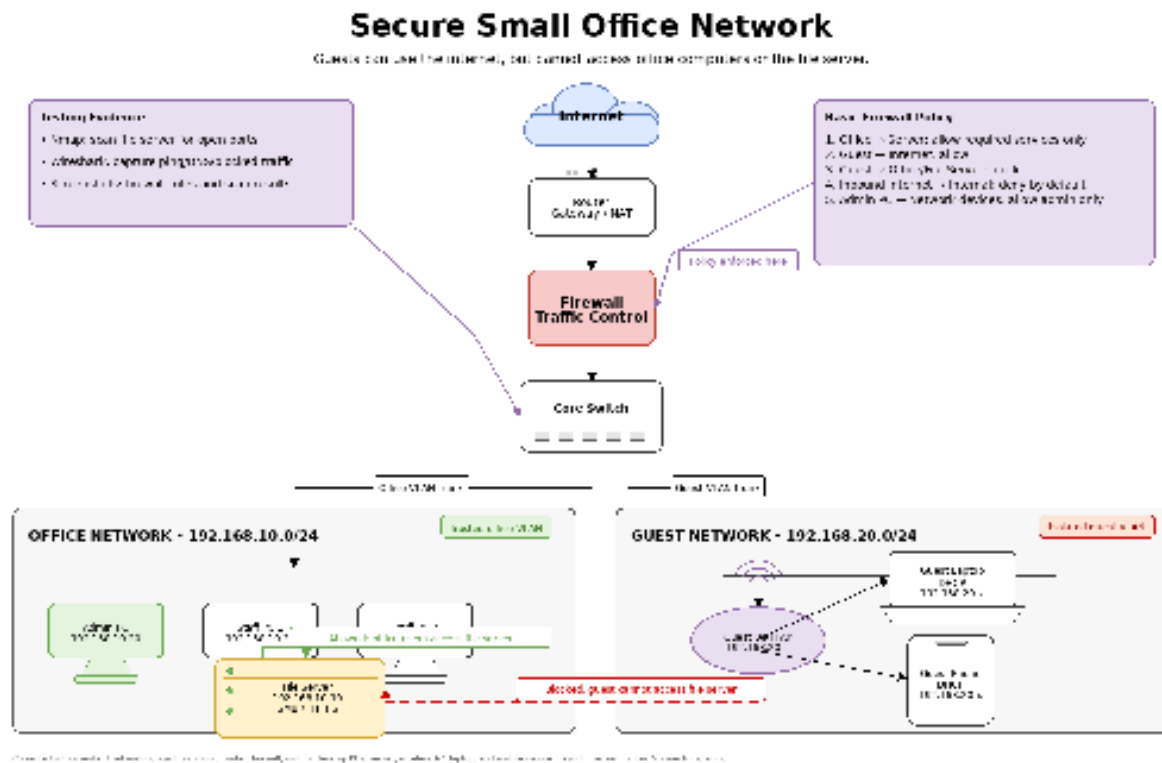


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Network Design

Clean diagram showing the office zone, guest zone, firewall policy and protected file server.



Guests can use the internet, but cannot access office computers or the file server.



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Devices and IP Address Plan

Trusted Office Zone

- Office Network: 192.168.10.0/24
- Office Gateway: 192.168.10.1
- Admin PC: 192.168.10.10
- Staff PC 1: 192.168.10.21
- Staff PC 2: 192.168.10.22
- File Server: 192.168.10.50

Guest Zone

- Guest Network: 192.168.20.0/24
- Guest Gateway: 192.168.20.1
- Guest Wi-Fi AP: 192.168.20.2
- Guest Devices: 192.168.20.100-200
- Guest devices are separated from internal office resources.

Key decision: the file server remains in the trusted office subnet while guest devices stay in a separate low-trust subnet.



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Basic Network Security Controls

1. Firewall Rules

- Office users can access the file server on required services.
- Guest-to-office traffic is blocked.
- Guest internet/DNS/DHCP is allowed.

2. Network Segmentation

- Office uses 192.168.10.0/24.
- Guest uses 192.168.20.0/24.
- Different zones reduce unauthorized access risk.

3. Access Control

- Admin PC is treated as the management workstation.
- Sensitive management access is limited.
- Strong passwords should be applied in a full deployment.

Core logic: Office to Server allowed | Guest to Office blocked | Guest to Internet allowed



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Methodology and Tools Used

Methodology

- Identified office requirements: staff need file-server access, guests need internet-only access.
- Designed the network zones and IP plan using draw.io.
- Built the topology in Cisco Packet Tracer with office and guest devices.
- Tested Admin PC access to the File Server.
- Tested Guest Laptop access to the same File Server.
- Used Simulation Mode to observe packet movement.

Tools Used

- draw.io: clean final network diagram.
- Cisco Packet Tracer: network simulation and ping tests.
- IP address plan: device and subnet consistency.
- Threat simulation notes: guest-device attack and response scenario.
- Project report materials: final explanation and evidence.

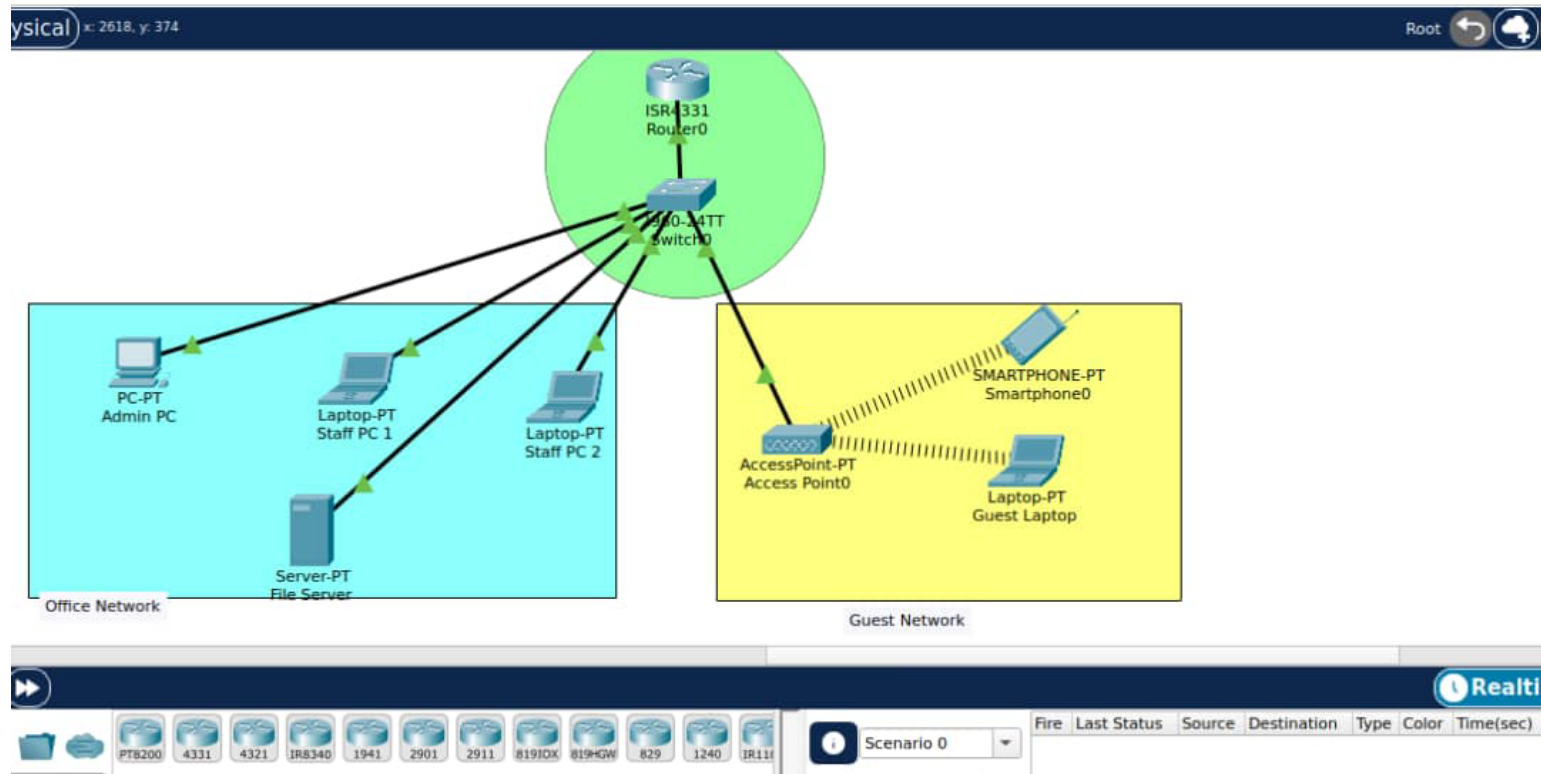


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Packet Tracer Topology Evidence

The topology was built in Cisco Packet Tracer with office and guest sections separated.



Topology showing Router0, Switch0, Admin PC, Staff PCs, File Server, AccessPoint0, Guest Laptop and Smartphone0.

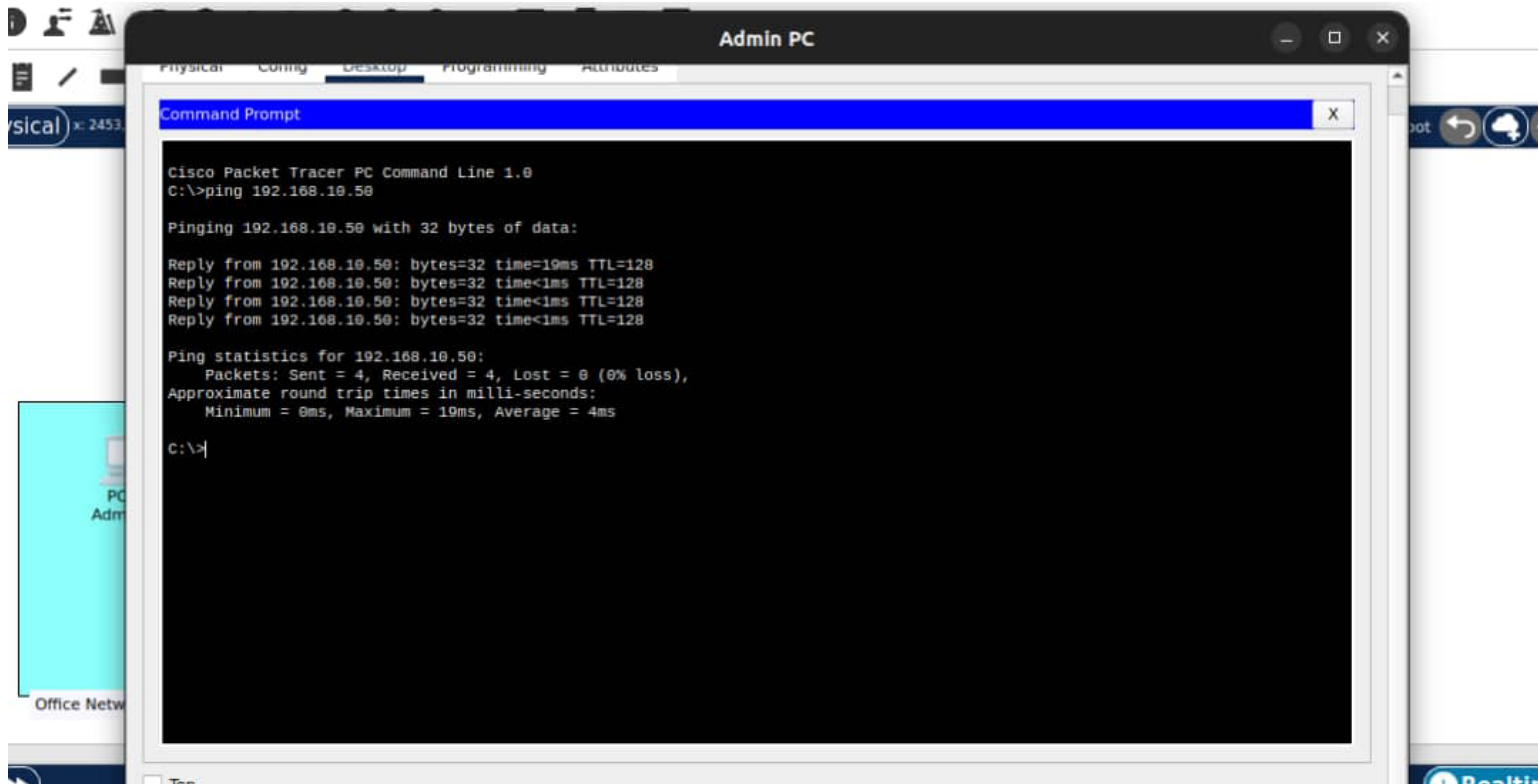


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Test 1: Admin PC to File Server Allowed

Trusted office traffic should be able to reach the internal File Server.



Admin PC ping to 192.168.10.50 succeeds with four replies and 0% packet loss.

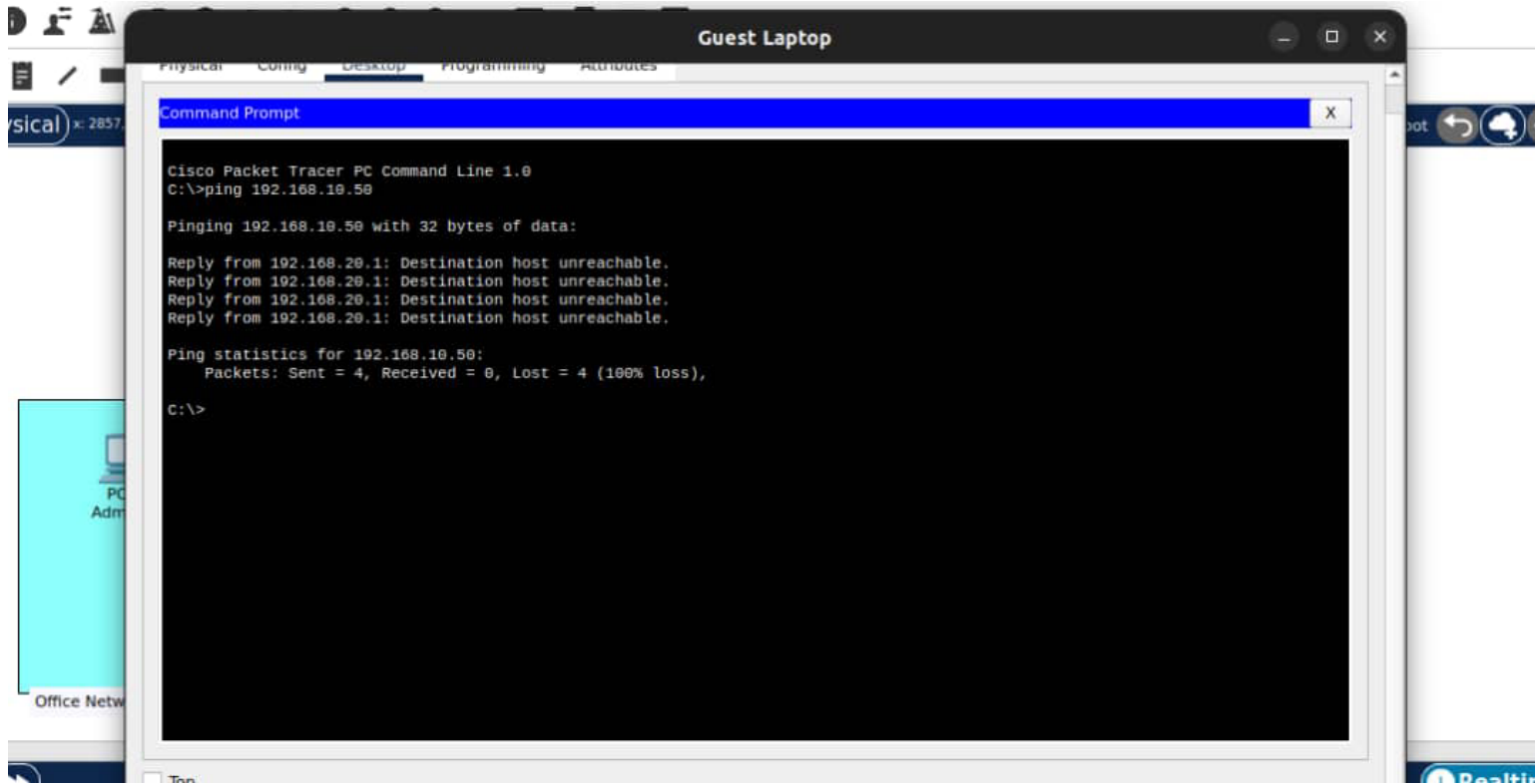


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Test 2: Guest Laptop to File Server Blocked

Guest traffic should not reach the office File Server.



Guest Laptop ping to 192.168.10.50 fails with Destination host unreachable and 100% packet loss.

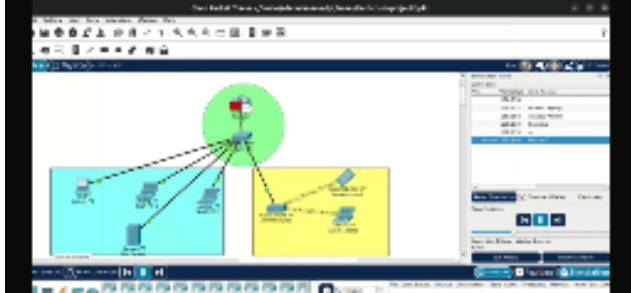


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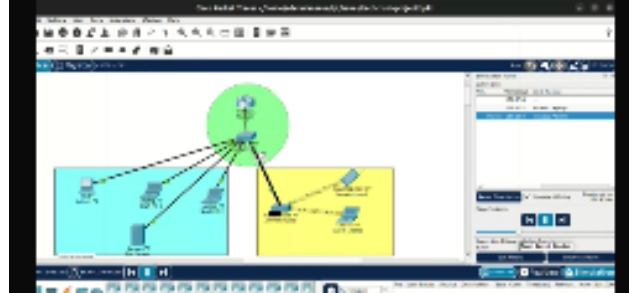
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Simulation Mode PDU/Event List Evidence

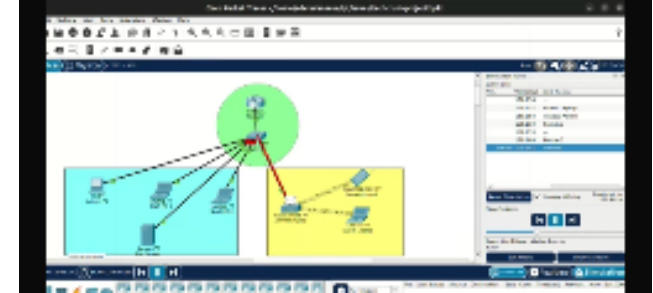
Packet Tracer Simulation Mode was used to show packet movement through the designed network.



Simulation overview and event list



Packet movement around the guest path



Blocked guest path toward office side

- The screenshots support the practical test, not just the static diagram.
- Traffic movement can be observed through the guest device, access point, switch and router path.
- The blocked guest path is consistent with the configured security goal.



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Simulated Threat and Response

Threat Scenario

- A guest connects to Wi-Fi and receives a 192.168.20.x address.
- The guest tries to discover or access the internal file server at 192.168.10.50.

Controls That Stop It

- Guest devices remain in 192.168.20.0/24.
- Firewall policy blocks Guest Network to Office Network traffic.
- The File Server remains inside 192.168.10.0/24.

Admin Response

- Identify the guest source device.
- Isolate it if needed.
- Check event information/logs.
- Confirm that no server access occurred.
- Document the incident and review rules.



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Results and Reflection

Key Results

- Required devices were included.
- Office and guest zones were separated.
- Admin PC reached the File Server.
- Guest Laptop failed to reach the server.
- Simulation Mode showed packet movement.

What We Learned

- Guest Wi-Fi must be separated from office resources.
- Security should protect the server while still allowing normal staff work.
- Controls must be tested, not only drawn.
- A secure design needs addressing, access rules and verification.

Future improvement: add firewall-rule screenshots, controlled Nmap scans, Wireshark captures and firewall logging in a full live-lab setup.



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Conclusion

- The project demonstrates how a small office network can be designed and defended with clear zones and basic access controls.
- The final design includes the required devices and separates office users from guest users.
- The Admin PC can reach the File Server at 192.168.10.50, confirming normal office access.
- The Guest Laptop cannot reach the same server, confirming guest isolation.
- The threat scenario and Simulation Mode evidence show how unauthorized guest access is stopped.



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References

- Cisco Systems. Cisco Packet Tracer documentation and networking simulation materials.
- Kurose, J. F., and Ross, K. W. Computer Networking: A Top-Down Approach.
- Stallings, W. Data and Computer Communications.
- diagrams.net / draw.io documentation for network diagramming.
- Course lecture materials and classroom notes on LAN design, VLANs, ACLs, wireless security and small office network security.
- Project materials reviewed: network diagram, IP address plan, Packet Tracer tests, Simulation Mode screenshots and threat-response notes.



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